



Clamp-On Micro Flow Sensor

NEW FD-X Series



**Monitor Micro Flow Rates
from Outside the Pipe**

Minimum instantaneous flow

0.1 mL/min

Minimum ejection amount

0.001 mL



FD-X Series

Monitor Low Flow Amounts from Outside the Pipe





Introducing KEYENCE's new series of clamp-on flow sensors that support minimal flows and narrow pipes

Clamp-On Micro Flow Sensor
FD-X Series **NEW**

Minimum display resolution
(Standard model)

Installable range	Instantaneous flow	Ejection amount
ø2.7 to ø14	0.1 mL/min	0.001 mL

Consider a clog detection type for simply checking clogs in the system

FD-XE Series [▶ P. 22](#)

Clamp-On Advantages

Zero contact with the liquid

No pressure loss
or contamination

Never clogs

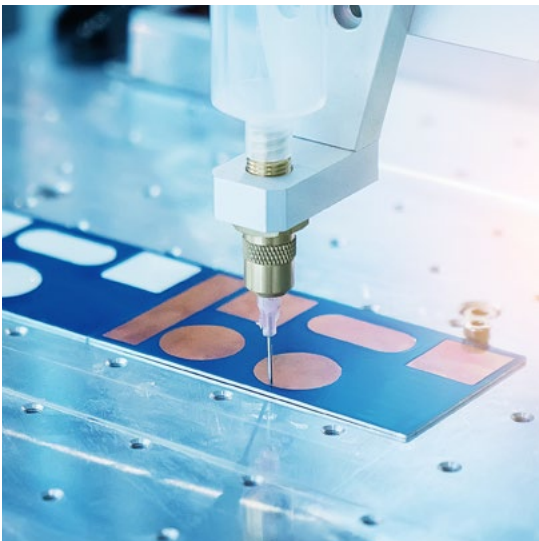
No need for
maintenance

No pipe modification

Easy
installation

Low Flow Monitoring + Clamp-On Design for Various Applications

Dispensing / Filling

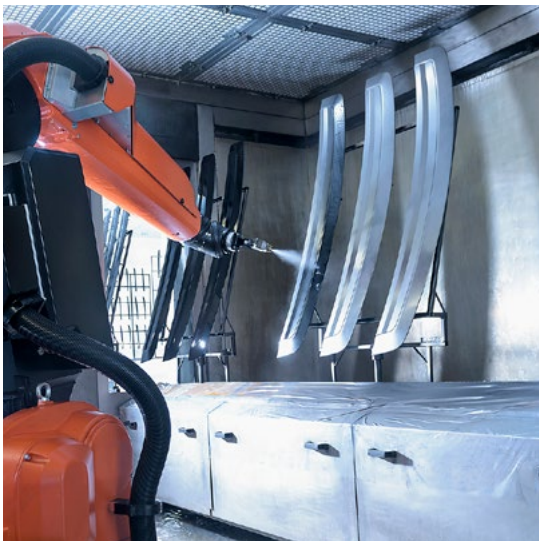


Non-Contact Design
Enables Monitoring of
Countless Liquids

The clamp-on design monitors flow from outside the pipe, making it suited for verifying coating amounts for grease, glue, and other materials where adherence to monitoring equipment is a concern. It can also verify fill amounts for food, medication, and other items where contact with monitoring equipment is to be avoided.

Example liquids			
Grease	Adhesive	Flux	Sealant
Anti-carbonising agent	Anti-rust agent	Resist fluid	Primer
FIPG	Silicone	Urethane	Varnish
Plating solution	Electrolyte	Solvent	Chemicals

Spraying / Coating / Circulation

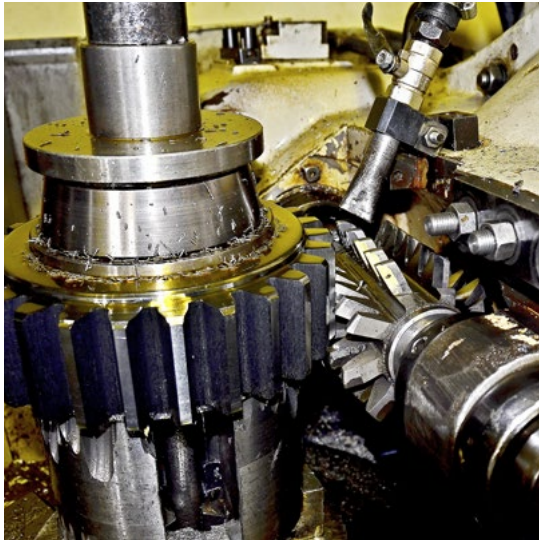


No Pressure Loss
Ensures Proper
Manufacturing Conditions

Typical flow sensors are placed within the flow channel, causing pressure losses and increasing the risk of varying manufacturing conditions. This makes it difficult to install conventional flow sensors on pre-existing equipment. A clamp-on flow sensor eliminates this risk and requires no maintenance.

Example liquids			
Mould release agent	Centre through coolant	Flocculant	Paint
Slurry	Alcohol	Beverages	Cooking oil
Additives	Ink		

Lubricating



No Downtime Needed,
Letting You Install Sensors in Places
You've Always Wanted to Monitor

The clamp-on design lets you install sensors in places you once thought were off-limits to flow sensors. This includes machines that could not be stopped for sensor installation and areas where oil or chemical exposure made conventional sensors deteriorate quickly.

Example liquids

Lubricating oil

Operating oil

Processing oil

Graphite

Cooling



Easy Installation
Lets You Instantly Add Sensors to
Complex Branched Piping

Clamp-on sensors do not require pipe modification, eliminating the need for pipe work during installation. Even pipes with complex branching can be equipped with sensors using just a single screwdriver. This ensures visualisation of flow even with tiny amounts of coolant or other liquids in pipe branches.

Example liquids

Cooling water

Pure water

Oil

Coolant

A Sensor Like None Before

Micro Flow Monitoring for Limitless Uses

Repeatability

$\pm 0.1\%$ of F.S.

Non-contact,
reliable measurements



Various types of

liquids

Monitor water, oil, and medications,
as well as highly viscous fluids



Compatible with many pipe materials

Supports plastic (soft/hard)
and metal piping



Space-saving compact size

Good for use in cramped
or complex areas

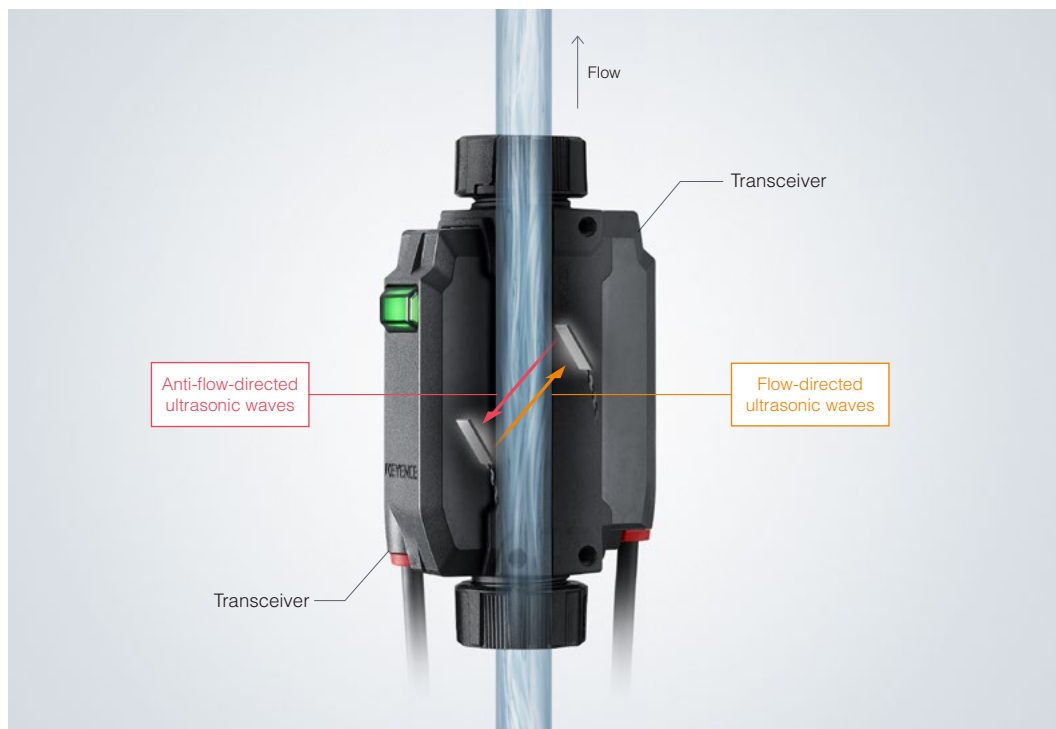


The FD-X Lineup, Supporting Pipe Sizes Between $\varnothing 2.7$ mm and $\varnothing 14$ mm

Three sensor heads and 13 clamp sets handle a wide variety of pipe sizes and materials
(See page 20 for details)

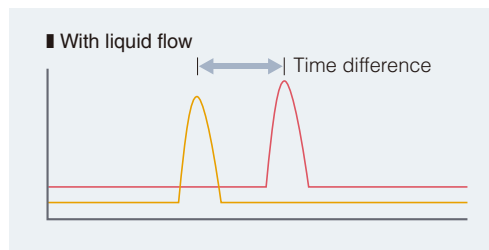


Repeatability of $\pm 0.1\%$ of F.S. Without Contacting the Liquid



Transmission Time Difference Method (DeltaTOF)

The sensor sends ultrasonic waves along, and against, the liquid flow direction. The faster the flow, the shorter the time required for the flow-directed waves to reach the other side, and the longer the time for the waves going in the opposite direction. Measuring this time difference ensures stable detection of flow, while eliminating the impact of environmental changes.



Superior Repeatability

Instantaneous flow: $\pm 0.1\%$ of F.S.

Ejection amount: ± 0.003 mL

Despite the external clamp-on design, the FD-X Series provides highly repeatable readings. It can be used to reliably monitor instantaneous flow, as well as fluid ejection rates.

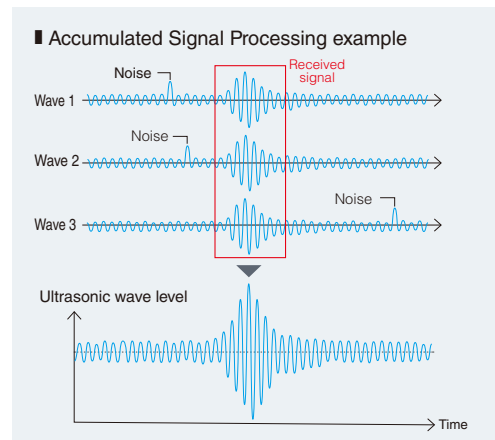
FD-XS8 repeatability
(example: $\phi 6$ pipe, 50 ms response time)

	Instantaneous flow	Ejection amount
Plastic piping /tubing	$\pm 0.1\%$ of F.S.	± 0.003 mL
Metal piping	$\pm 0.3\%$ of F.S.	± 0.008 mL

See page 26

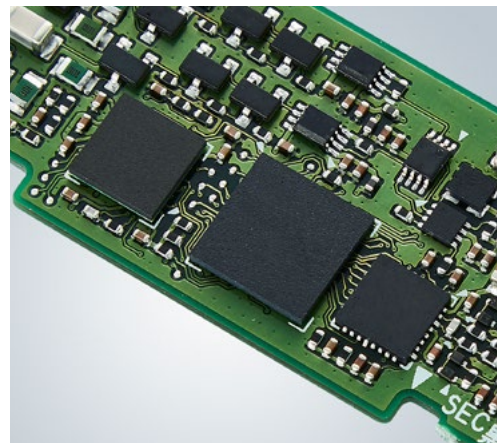
Accumulated Signal Processing Cuts Out Noise

High-speed accumulated signal processing all but eliminates the impact of noise. Only looking at one signal makes the impact of noise high. That is why the FD-X Series combines multiple signals to consistently recognise only the correct waveform.



Triple-Core Parallel Processing for Higher Response Speeds

The FD-X Series features an impressive 50 ms response time, while also maintaining stable and consistent readings. This is made possible by the FPGA circuit and two micro-controllers, which enable triple-core parallel processing.



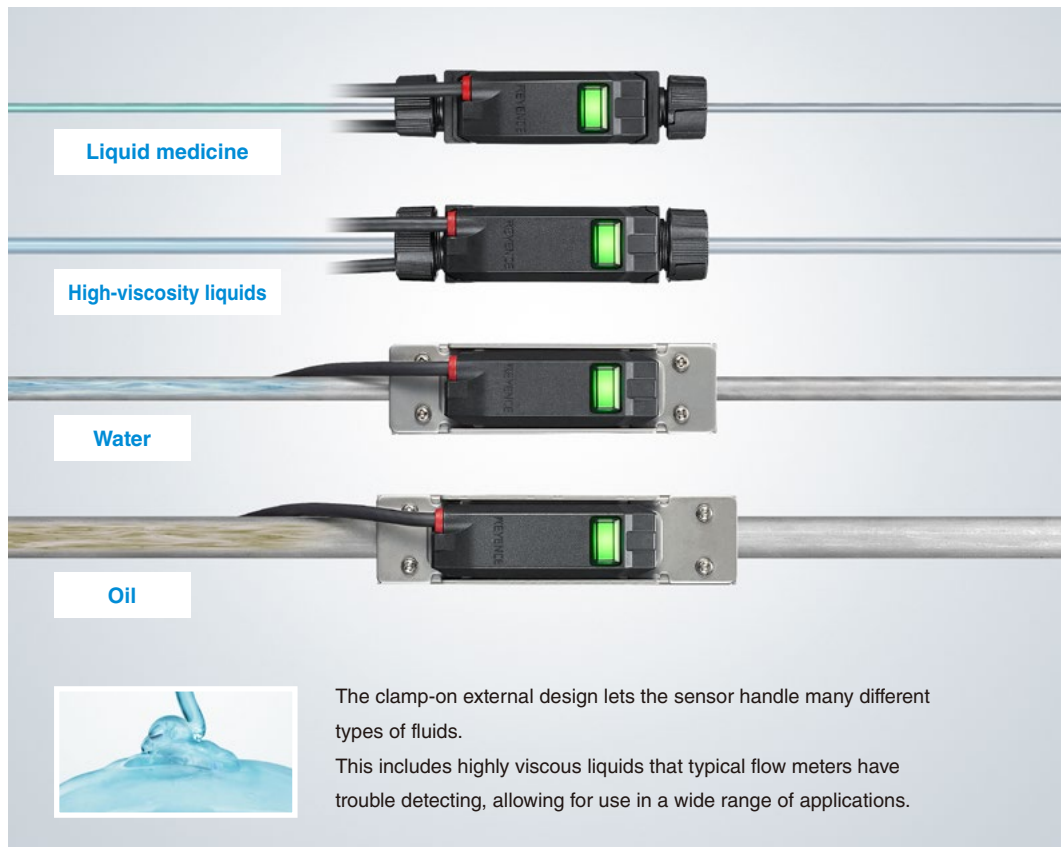
High Repeatability with Easy Installation

For plastic piping/tubing, simply clamp-on using your hands. For metal pipes, just tighten the screws, and installation is complete.

Along with this, there is no sensor greasing required, which is common for conventional ultrasonic flow meters.

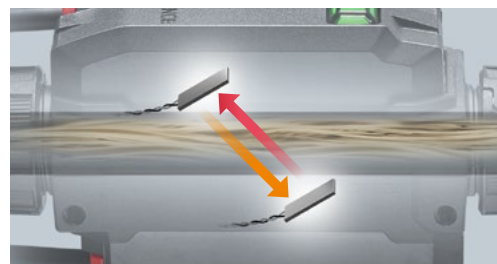


Stably Monitor Countless Fluids



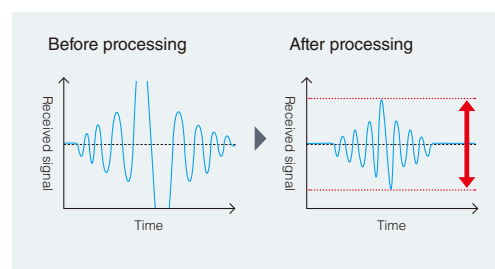
Thrubeam Structure Enables Higher Power

High-power elements are placed opposite of each other, similar to a thrubeam sensor, boosting signal reception. This lets the FD-X Series handle a plethora of fluids, including high-turbidity paints and high-viscosity grease.

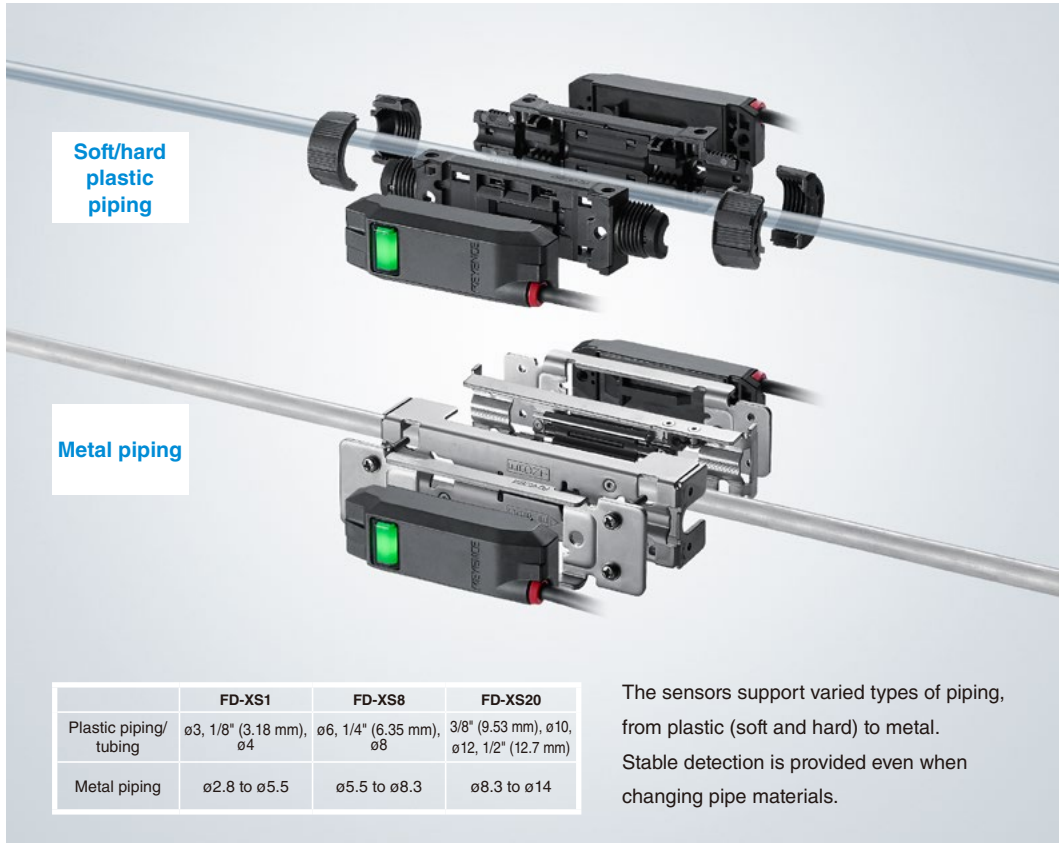


3000x High Dynamic Range

Every fluid has a different ultrasonic wave attenuation rate. The FD-X Series controls transmission and reception strength to maintain a constant signal reception level. This lets it stably monitor nearly all types of liquids.



Compatible with a Variety of Piping

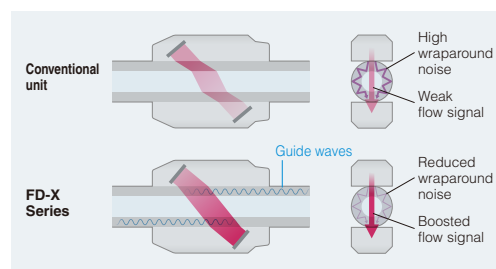


	FD-XS1	FD-XS8	FD-XS20
Plastic piping/ tubing	ø3, 1/8" (3.18 mm), ø4	ø6, 1/4" (6.35 mm), ø8	3/8" (9.53 mm), ø10, ø12, 1/2" (12.7 mm)
Metal piping	ø2.8 to ø5.5	ø5.5 to ø8.3	ø8.3 to ø14

The sensors support varied types of piping, from plastic (soft and hard) to metal. Stable detection is provided even when changing pipe materials.

Guided Waves Detect Flow in Thin Metal Piping World first

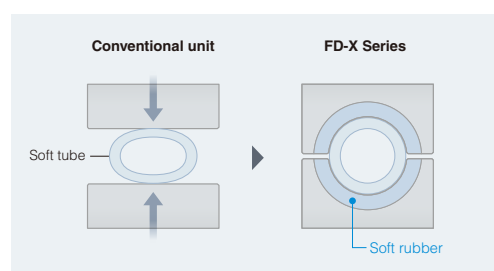
Guide waves are used to reduce the effects of wraparound noise generated in thin metal piping. Creating resonance on the pipe walls themselves improves wave permeation through the flowing liquid, allowing for stable signal transmission.



Uniform Clamping Around Plastic Piping/Tubing

When detecting flow through soft plastic tubes, it is vital to not crush the tube with the clamp set.

The FD-X Series uses soft rubber to clamp uniformly around the tube diameter, and prevent deformation.



Small Body for Space-Saving Installation

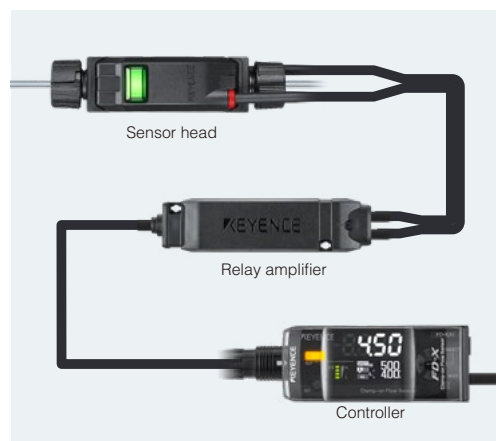


Devices with small flow amounts often use small-bore pipes placed in close proximity to each other. When inserting sensors into these scenarios, the size of the sensor becomes an important factor. To address this need, the FD-X Series features a sensor head that has been miniaturised as much as possible to eliminate concerns about installation space.



Separate Construction To Reduce Sensor Head Size

The sensor head design features a separate relay amplifier to reduce the size of the sensor head. This also reduces heat generation in the head, preventing it from changing the properties of the flowing liquid.



IP68G-Compliant Enclosure

The sensor head and relay amplifier both feature an IP68G (JIS C0920) enclosure rating. This allows for reliable operation in applications where oil is present, such as managing lubricant for presses.

IP68G

- No ingress of oil drops/droplets from any direction.
- Protected against the effects of continuous immersion in water.
- No ingress of dust.

* Testing at internal-standard conditions confirms no damage to devices.



Plastic Sensor Head Case for Handling Liquid Medication

The sensor head utilises a plastic housing, making it a valid solution for medical environments where metallic components are not allowed. The case uses a carbon fibre-reinforced polymer (CFRP) for superior chemical resistance and strength.



Dual Displays to Show a Range of Information

The controller display panel features a bright white LED display and an organic LED screen. This makes numbers easily visible, as well as allowing it to display graphs on the OLED screen. Changes over time and trends can be viewed using just the sensor alone.



The OLED screen provides a great variety of displays.



Graph displaying the liquid ejection values



Timeline display of the instantaneous flow rate

Full-Time Recording for Easy Analysis and Management



The FD-X Series continually saves pertinent data in its main unit without using a storage device.

This data can be viewed on the controller and sent via USB to a computer.

Saved data

Instantaneous flow

Accumulated flow

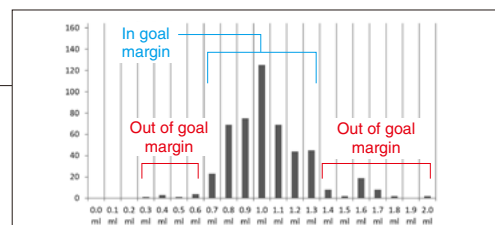
Ejection amount

On/off statuses, assorted alarms

■ Useful Applications

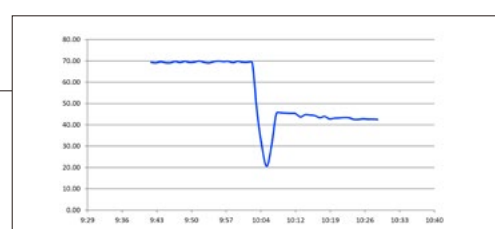
1. Liquid ejection/application management (histograms)

The sensor automatically saves ejection amount data after every instance. Outputting this data via USB lets you visualise the distribution state of ejection/application compared to your goals. This helps to identify abnormalities or issues in the process.



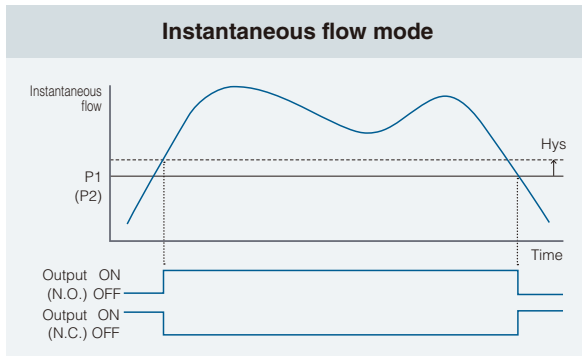
2. Instantaneous/accumulated flow management (trend graphs)

Peak/bottom instantaneous and accumulated flow values are automatically saved at regular intervals from the moment power is turned on. This helps with trend management, usage quantity management, and much more.



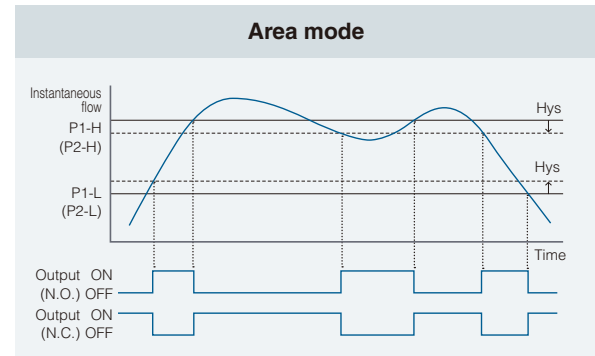
Selectable Detection Modes

Want to know if the instantaneous flow rate is dropping



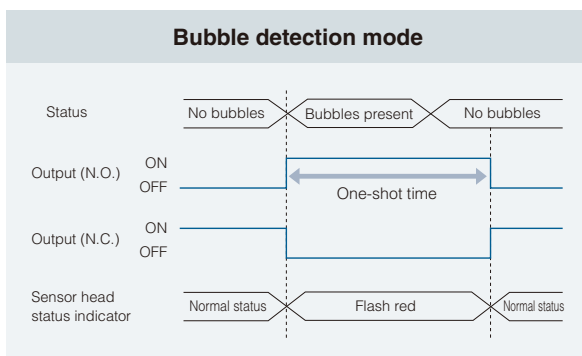
The output changes when the instantaneous flow rate crosses a set threshold.

Want to know if the instantaneous flow rate is outside a reasonable range



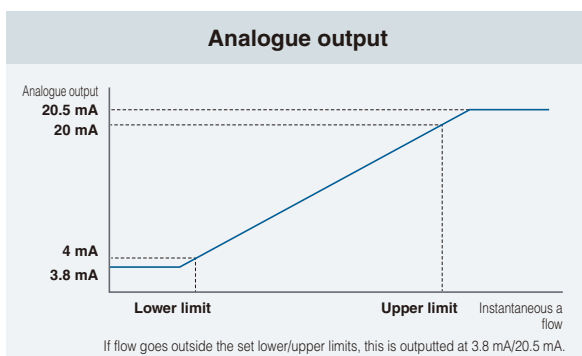
The output changes if flow goes outside the acceptable range.

Want to know if there are air bubbles in the fluid



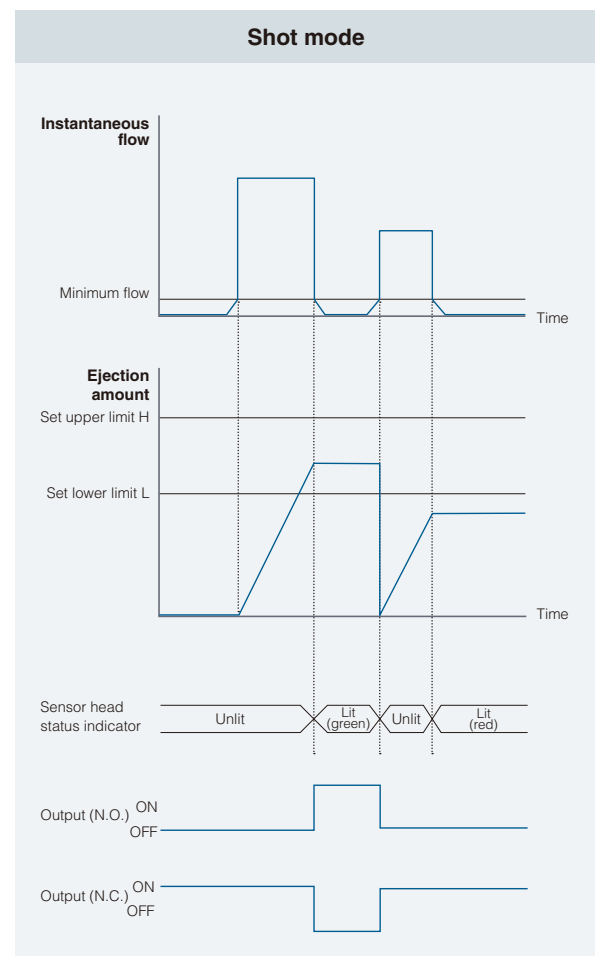
This mode detects bubbles in the piping and provides a one-shot output.

Want to display instantaneous flow-rate data on an external device



This signal is output from 4-20 mA or 0-20 mA depending on the set lower/upper limits. (Graph above is for 4-20 mA)

Want to see if the amount of liquid ejected is within the acceptable range



The sensor can detect each shot for application/ejection and automatically reset these values (auto-detect mode).

Easier and More Convenient than Ever

Press the SET Button for Easy Setup



Prior to the FD-X Series, set values needed to be directly input into the main unit. Now, all it takes is the press of a button to quickly and properly calibrate an appropriate set point.

Fully Automatic Tuning



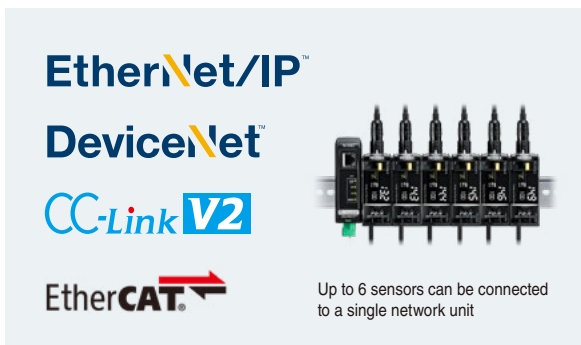
The sensor automatically chooses an appropriate set point by sampling flow values while SET is pressed.

Target Calibration



Enter values measured with an electronic balance or similar device, and the sensor will correct the readings based on the accumulated data gathered while SET+MODE are pressed.

Network Compatibility (NU Series)



The FD-X Series can connect to the NU Series, which supports assorted open field networks.

* Contact your local KEYENCE representative for more details.

IO-Link Compatibility

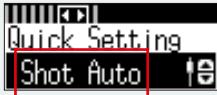


The controller also supports IO-Link communication. The device can output control signals, as well as data like instantaneous flow and ejection rates. This is useful for predictive maintenance and continuous monitoring.

Quick Setting Function NEW

Three selectable modes

Shot Auto	Standard discharge
Shot Manual	High-accuracy discharge
Minimum Flow	Minimal instantaneous flow rate



When "Shot Auto" is selected *

Setting (excerpt)	Initial value	Setting value
CH.1	Instantaneous flow mode	Shot mode
Input 2	Off	Zero shift input
Response time	500 ms	50 ms
Zero cut flow rate	20 mL/min	10 mL/min

* With sensor head FD-XS1

Recommended settings are provided for three typical applications. Selecting any of these modes will change the settings from the default to the optimal parameter depending on the purpose of use, enabling immediate use without the need to configure complicated settings.

Select from DIN/Panel Mount Types



The controller comes in DIN rail mount and panel mount styles. Select the most suitable controller for your machine's layout.

Large Status Indicator Light



The large status indicator lets you intuitively see the state of the sensor output. Put it in predictive maintenance mode, and the light pattern will indicate when maintenance is necessary before it is too late.

Simulation Mode



Simulated instantaneous and ejection flow amounts can be input into the device, making debugging before installation easier. Both the on/off control output and the analogue output can be tested.

Status Monitoring Mode

Conventional unit




A single glance won't tell you if the flow rate is dropping.

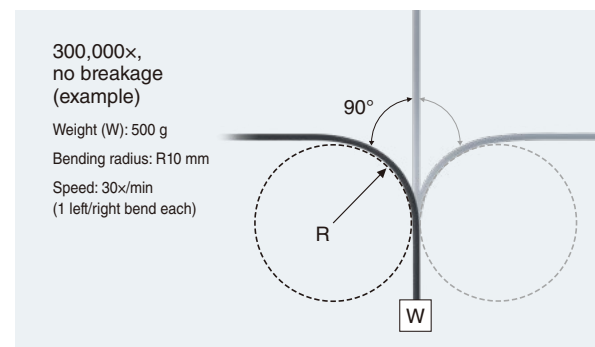
Status Monitor Mode




Setting normal operation as 100% lets you quickly discover status changes.

Saving the normal status as a baseline (100%) lets you check relative values to see if the status has changed at all. This simplified display is helpful for determining maintenance issues early on.

Hi-Flex Robotic Cable



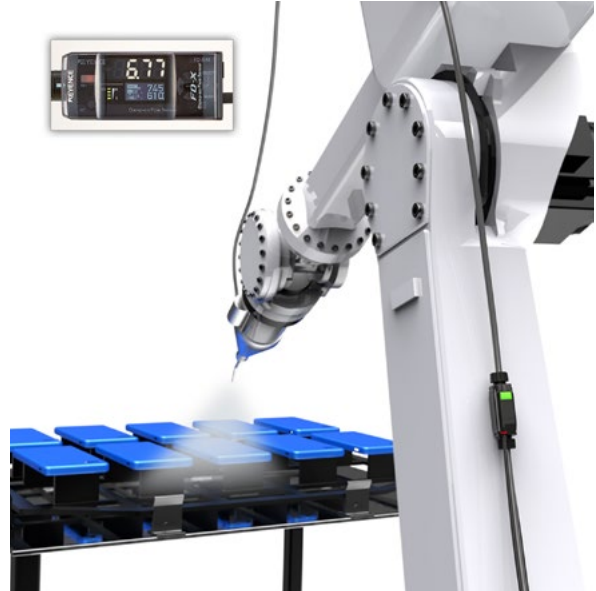
A high-flex cable links the relay amplifier with the controller. This makes it usable even on moving machinery.

Applications



Checking Dispenser Application Amounts

Ideal when using sensitive liquids, the sensor head does not generate excessive heat while monitoring liquid flow from outside the pipe. This is perfect for quality control and management of liquid amounts used.



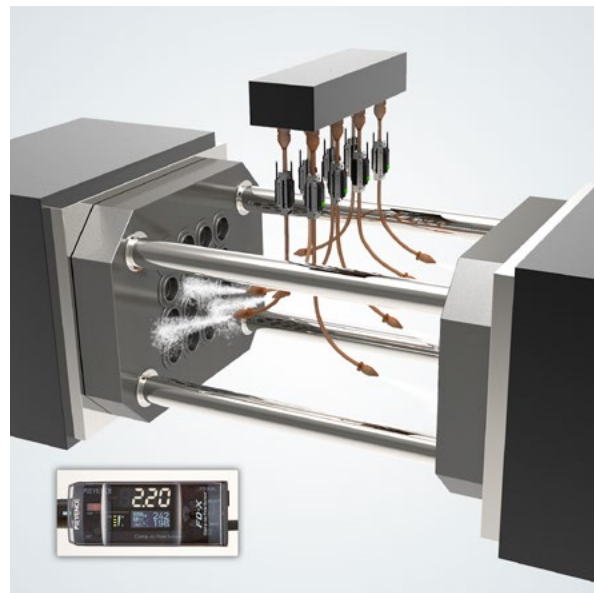
Checking Spray Rates for Robots

Monitor the flow of coating liquids without the sensor directly coming into contact with them, eliminating the need for periodic maintenance.



Checking Lubricant Flow

The FD-X Series offers a high enclosure ratings, which allows it to operate in harsh environments. This includes industrial presses with heavy lubricant flow.



Checking Application Rates of Mould Release Fluid

The sensor provides reliable measurement without cutting into piping. The small sensor head also makes it installable in areas with lots of tubing in one place.



Checking Coolant Flow Rates

The small-sized FD-X Series can monitor minute amounts of flow, making it perfect for managing flow rates in complex, branched coolant tubes inside casting-machine moulds.



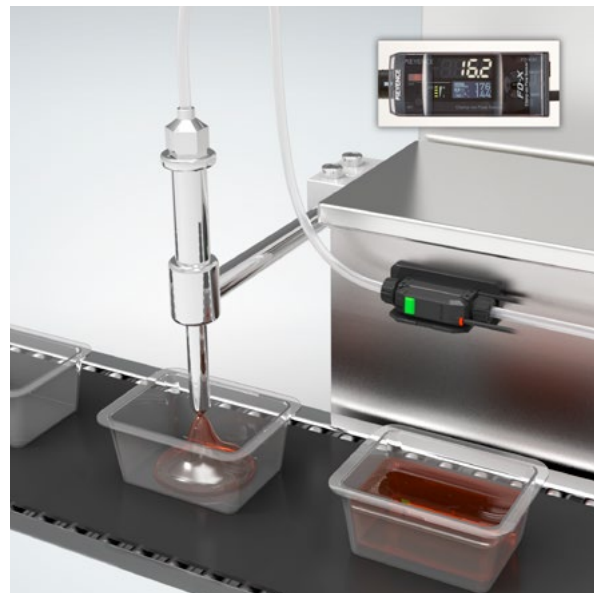
Checking Flow Rates on Liquid Drug Injectors

The sensor monitors flow from outside the pipe, eliminating the need to worry about chemical resistance. The high repeatability lets you use it to manage the amount of injected liquid.



Checking Sealant Application Rates

The clamp-on design allows for reliable readings of highly viscous sealant material, without causing drops in pressure,



Checking Fill Amounts for Food

Contamination is not a concern using the non-contact FD-X Series. This makes it ideal for management of food products.

Selection Guide FD-X Series (Standard)

STEP 1 Clamp Set and Sensor Head Selection

For plastic piping/tubing



1. Select the clamp set based on the outer diameter of the piping.

2. Select the corresponding sensor head.

Target pipe diameter		A Clamp set		
Pipe outer diameter*	Installable range	Appearance	Model	Weight
ø3	ø2.7 to 3.7		FD-XC1R1	Approx. 50 g
1/8" (3.18 mm)			FD-XC1R2	Approx. 50 g
ø4	ø3.5 to 4.5		FD-XC8R1	Approx. 55 g
ø6	ø5.5 to 6.5		FD-XC8R2	Approx. 60 g
1/4" (6.35 mm)	ø5.9 to 6.9		FD-XC8R3	Approx. 60 g
ø8	ø7.5 to 8.5		FD-XC20R1	Approx. 75 g
3/8" (9.53 mm)	ø9.0 to 10.0		FD-XC20R2	Approx. 80 g
ø10	ø9.5 to 10.5		FD-XC20R3	Approx. 80 g
ø12	ø11.5 to 12.5		FD-XC20R4	Approx. 80 g
1/2" (12.7 mm)	ø12.2 to 13.2			

B Sensor head			Rated flow range
Appearance	Model	Weight	
	FD-XS1	Approx. 230 g	0 to 1000 mL/min
	FD-XS8	Approx. 250 g	0 to 3000 mL/min 0 to 8000 mL/min
	FD-XS20	Approx. 260 g	0 to 15 L/min 0 to 20 L/min

* The dimensions in inch are not Nominal dimensions B of JIS/ANSI standards. 1 inch = 25.4 mm

For metal piping



1. Select the clamp set based on the outer diameter of the piping.

2. Select the corresponding sensor head.

Target pipe diameter			A Clamp set		
Pipe outer diameter*	A name	Installable range	Appearance	Model	Weight
ø3	—	ø2.8 to 5.5 mm		FD-XC1M	Approx. 190 g
1/8" (3.18 mm)	—				
ø4	—	ø5.5 to 8.3 mm		FD-XC8M	Approx. 210 g
ø6	—				
1/4" (6.35 mm)	—				
ø8	—	ø8.3 to 10.8 mm		FD-XC20M1	Approx. 240 g
3/8" (9.53 mm)	—				
ø10	—				
ø10.5	6 A				
ø12	—	ø10.8 to 14 mm		FD-XC20M2	Approx. 250 g
1/2" (12.7 mm)	—				
ø13.8	8 A				

B Sensor head			Rated flow range
Appearance	Model	Weight	
	FD-XS1	Approx. 230 g	0 to 1000 mL/min
	FD-XS8	Approx. 250 g	0 to 3000 mL/min 0 to 8000 mL/min
	FD-XS20	Approx. 260 g	0 to 15 L/min 0 to 20 L/min

* The dimensions in inch are not Nominal dimensions B of JIS/ANSI standards. 1 inch = 25.4 mm

STEP 2 Controller Selection

Controllers

Type	Appearance	Model	Control output	External input	Analogue current output	Network Compatibility	Cable	Weight (with cable)
DIN-rail mount type, main unit		FD-XA1	2 outputs (selectable NPN/PNP)	2 inputs	1 output	IO-Link	7-core loose wires cable, 2 m	Approx. 210 g
DIN-rail mount type, expansion unit Up to 7 expansion units per main unit		FD-XA2			—	NU Series • EtherNet/IP™ • CC-Link • DeviceNet™ • EtherCAT®	4-core loose wires cable, 2 m	Approx. 180 g
Panel mount type, main unit		FD-XA5			1 output	IO-Link	7-core loose wires connector cable included, 2 m	Approx. 210 g

Network Communication Unit,
Multi-Output Unit
(select as needed)

Contact your local KEYENCE
representative for more details

Network
Communication Unit
NU Series



Multi-Output Unit
FS-MC8N/MC8P
Controller settings
can be saved and
written



STEP 3 Optional Parts Selection (if needed)

Installation

Type	Appearance	Model	For use with	Description	Weight
Securing bracket for plastic clamp set		OP-88294	FD-XC1R1/XC1R2 FD-XC8R1/XC8R2/XC8R3 FD-XC20R1/XC20R2/XC20R3/ XC20R4	For securing the plastic clamp set to a jig, etc. Use when clamping to a soft plastic tube.	Approx. 55 g
Metal clamp set mounting bracket		OP-88297	FD-XC1M/XC8M	For securing the metal clamp set to a jig, etc. Use with metal piping with an outer diameter of ø8.3 mm or less if heavy vibration or shocks occur in the installation area.	Approx. 60 g
PEEK screw set		OP-88295	FD-XC1R1/XC1R2 FD-XC8R1/XC8R2/XC8R3 FD-XC20R1/XC20R2/XC20R3/ XC20R4/OP-88294	Use if the chemical resistance of the SUS screws included with the plastic clamp sets (FD-XCxxRx) or plastic clamp securing brackets (OP-88294) is a concern.	Approx. 3 g
DIN securing bracket (for main unit)		OP-88311	FD-XA1	Allows attachment without a DIN rail.	Approx. 15 g
End unit (for expansion)		OP-26751	FD-XA1/XA2	Secure main and expansion units when mounted together on a DIN rail. Always use when connecting multiple units. (Pack of 2)	Approx. 15 g

Wiring

Type	Appearance	Model	For use with	Description	Weight
Sensor head-controller extension cable, 2 m		OP-88292	FD-XS1/XS8/XS20	A cable that further extends the 2 m cable between the sensor head's relay amp and the controller. Connectors are on both ends.	Approx. 110 g
Sensor head-controller extension cable, 5 m		OP-88293		* The cable between the relay amp and controller can be extended up to 12 m long.	Approx. 240 g
Loose wires/M12 adapter		OP-88296	FD-XA1/XA5 cables, or cables with cross sectional area 0.14 to 0.34 mm² and outer diameter ø3.5 to 6 mm	A connector that converts loose wires to a M12 4-pin connector. Useful for connecting to IO-Link compatible master units.	Approx. 12 g

Clamp-on to detect clogs in the system

Compatible pipe diameters

ø2.7 mm to ø14 mm



Detect clogs from outside the pipe



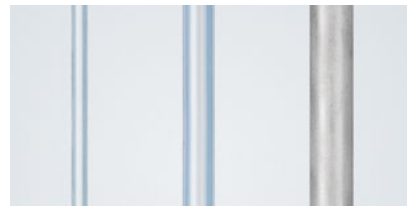
Clogged!



Works with a range of liquids and pipes

Monitor **water**, **oils**, and **medications**, as well as highly **viscous fluids**

Supports **plastic (soft/hard)** and **metal piping**



Perfect for use where clogging is a concern



Check for clogging during sealant application

Clogging of viscous sealant materials can be accurately detected from outside the pipe without any loss of pressure.



Check for clogging in coating robots

The FD-X Series will not clog even when monitoring viscous liquids as there is no physical contact with the liquid.



Check for clogging of lubricating oil

The excellent environmental resistance ensures problem-free usability even in extreme environments such as when managing press lubricating oil.



Check for clogging in coolant lines

The impressively small sensor head eliminates concerns when inserting in intricately branching coolant pipes found inside moulding machines.

Selection Guide FD-XE Series (For clog detection)

STEP1 Clamp Set and Sensor Head Selection

For plastic piping/tubing



1. Select the clamp set based on the outer diameter of the piping.

2. Select the corresponding sensor head.

Target pipe diameter		A Clamp set			B Sensor head			Maximum rated flow
Pipe outer diameter*	Installable range	Appearance	Model	Weight	Appearance	Model	Weight	
ø3	ø2.7 to 3.7		FD-XC1R1	Approx. 50 g		FD-XS1E	Approx. 230 g	1000 mL/min
1/8" (3.18 mm)			FD-XC1R2	Approx. 50 g				
ø4	ø3.5 to 4.5		FD-XC8R1	Approx. 55 g		FD-XS8E	Approx. 250 g	3000 mL/min
ø6	ø5.5 to 6.5		FD-XC8R2	Approx. 60 g				8000 mL/min
1/4" (6.35 mm)	ø5.9 to 6.9		FD-XC8R3	Approx. 60 g				8000 mL/min
ø8	ø7.5 to 8.5		FD-XC20R1	Approx. 75 g		FD-XS20E	Approx. 260 g	15 L/min
3/8" (9.53 mm)	ø9.0 to 10.0		FD-XC20R2	Approx. 80 g				20 L/min
ø10	ø9.5 to 10.5		FD-XC20R3	Approx. 80 g				
ø12	ø11.5 to 12.5		FD-XC20R4	Approx. 80 g				
1/2" (12.7 mm)	ø12.2 to 13.2							

* The dimensions in inch are not Nominal dimensions B of JIS/ANSI standards. 1 inch = 25.4 mm

For metal piping



1. Select the clamp set based on the outer diameter of the piping.

2. Select the corresponding sensor head.

Target pipe diameter			A Clamp set			B Sensor head			Maximum rated flow
Pipe outer diameter*	A name	Installable range	Appearance	Model	Weight	Appearance	Model	Weight	
ø3	—	ø2.8 to 5.5 mm		FD-XC1M	Approx. 190 g		FD-XS1E	Approx. 230 g	1000 mL/min
1/8" (3.18 mm)	—								
ø4	—								
ø6	—	ø5.5 to 8.3 mm		FD-XC8M	Approx. 210 g		FD-XS8E	Approx. 250 g	3000 mL/min
1/4" (6.35 mm)	—								8000 mL/min
ø8	—								
3/8" (9.53 mm)	—	ø8.3 to 10.8 mm		FD-XC20M1	Approx. 240 g		FD-XS20E	Approx. 260 g	15 L/min
ø10	—								20 L/min
ø10.5	6 A								
ø12	—								
1/2" (12.7 mm)	—								
ø13.8	8 A	ø10.8 to 14 mm		FD-XC20M2	Approx. 250 g		FD-XS20E	Approx. 260 g	20 L/min

* The dimensions in inch are not Nominal dimensions B of JIS/ANSI standards. 1 inch = 25.4 mm

STEP2 Controller Selection

Controllers

Type	Appearance	Model	Control output	External input	Analogue current output	Network Compatibility	Cable	Weight (with cable)
DIN-rail mount type, main unit		FD-XA1E	Up to 2 outputs (selectable NPN/PNP)	Up to 1 input	—	—	4-core loose wires cable, 2 m	Approx. 210 g
DIN-rail mount type, expansion unit Up to 7 expansion units per main unit		FD-XA2E			—	NU Series • EtherNet/IP™ • CC-Link • DeviceNet™ • EtherCAT®	2-core loose wires cable, 2 m	Approx. 180 g
Panel mount type, main unit		FD-XA5E			—	—	4-core loose wires connector cable included, 2 m	Approx. 210 g

Network Communication Unit, Multi-Output Unit (select as needed)

Contact your local KEYENCE representative for more details

Network Communication Unit
NU Series



Multi-Output Unit
FS-MC8N/MC8P
Controller settings can be saved and written



STEP3 Optional Parts Selection (if needed)

Installation

Type	Appearance	Model	For use with	Description	Weight
Securing bracket for plastic clamp set		OP-88294	FD-XC1R1/XC1R2 FD-XC8R1/XC8R2/XC8R3 FD-XC20R1/XC20R2/XC20R3/XC20R4	For securing the plastic clamp set to a jig, etc. Use when clamping to a soft plastic tube.	Approx. 55 g
Metal clamp set mounting bracket		OP-88297	FD-XC1M/XC8M	For securing the metal clamp set to a jig, etc. Use with metal piping with an outer diameter of ø8.3 mm or less if heavy vibration or shocks occur in the installation area.	Approx. 60 g
PEEK screw set		OP-88295	FD-XC1R1/XC1R2 FD-XC8R1/XC8R2/XC8R3 FD-XC20R1/XC20R2/XC20R3/XC20R4/OP-88294	Use if the chemical resistance of the SUS screws included with the plastic clamp sets (FD-XCxx) or plastic clamp securing brackets (OP-88294) is a concern.	Approx. 3 g
DIN securing bracket (for main unit)		OP-88311	FD-XA1E	Allows attachment without a DIN rail.	Approx. 15 g
End unit (for expansion)		OP-26751	FD-XA1E/XA2E	Secure main and expansion units when mounted together on a DIN rail. Always use when connecting multiple units. (Pack of 2)	Approx. 15 g

Wiring

Type	Appearance	Model	For use with	Description	Weight
Sensor head-controller extension cable, 2 m		OP-88292	FD-XS1E/XS8E/XS20E	A cable that further extends the 2 m cable between the sensor head's relay amp and the controller. Connectors are on both ends. * The cable between the relay amp and controller can be extended up to 12 m long.	Approx. 110 g
Sensor head-controller extension cable, 5 m		OP-88293			Approx. 240 g

Specifications



■ FD-X (Standard)

Sensor head model				FD-XS1			FD-XS8			FD-XS20				
Supported pipe materials				Metal pipes, Plastic pipes (soft/hard)* ¹										
Supported fluids				Liquids (water, oil, adhesive, grease, chemical solution, etc.)* ¹										
Supported fluid temperature (Pipe surface temperature)				0°C (no freezing on the pipe surface) to 100°C										
Supported diameter	Plastic pipe/tube attachment	Clamp set model	FD-XC1R1		FD-XC1R2	FD-XC8R1	FD-XC8R2	FD-XC8R3	FD-XC20R1	FD-XC20R2		FD-XC20R3	FD-XC20R4	
		Outer diameter of pipe	ø3	1/8" (3.18 mm)	ø4	ø6	1/4" (6.35 mm)	ø8	3/8" (9.53 mm)	ø10		ø12	1/2" (12.7 mm)	
		Attachable range	ø2.7 to 3.7		ø3.5 to 4.5	ø5.5 to 6.5	ø5.9 to 6.9	ø7.5 to 8.5	ø9.0 to 10.0	ø9.5 to 10.5		ø11.5 to 12.5	ø12.2 to 13.2	
	Metal pipe attachment	Clamp set model	FD-XC1M			FD-XC8M			FD-XC20M1			FD-XC20M2		
		Outer diameter of pipe	ø3	1/8" (3.18 mm)	ø4	ø6	1/4" (6.35 mm)	ø8	3/8" (9.53 mm)	ø10	ø10.5	ø12	1/2" (12.7 mm) ø13.8	
		A designation	—	—	—	—	—	—	—	—	6 A	—	— 8 A	
		Attachable range	ø2.8 to 5.5			ø5.5 to 8.3			ø8.3 to 10.8			ø10.8 to 14		
	Rated flow range				0 to 1000 mL/min			0 to 3000 mL/min	0 to 8000 mL/min	0 to 15.00 L/min		0 to 20.00 L/min		
Zero cut flow rate* ² (variable, default)				20 mL/min			40 mL/min			0.10 L/min (standard), 40 mL/min (high resolution* ⁶)				
Display resolution (Displayed on controller)		Instantaneous flow rate		0.1/1/10 mL/min					0.001/0.01/0.1 L/min (standard), 0.1/1/10 mL/min (high resolution* ⁶)					
		Shot amount		0.001/0.01/0.1/1 mL					0.001/0.01/0.1 L (standard), 0.001/0.01/0.1/1 mL (high resolution* ⁶)					
Repeat-ability* ³	Plastic pipe/tube attachment	Response time: 50 ms* ⁴	F.S.	±0.6%				±0.1%						
		Instantaneous flow rate	±6 mL/min				±3 mL/min		±8 mL/min		±15 mL/min		±20 mL/min	
		Response time: 500 ms	Instantaneous flow rate	±1.9 mL/min		±1.0 mL/min		±2.6 mL/min		±4.7 mL/min		±6.3 mL/min		
	Metal pipe attachment	Response time: 50 ms* ⁴	F.S.	±1%		±0.3%		±0.15%						
		Instantaneous flow rate	±10 mL/min		±9 mL/min		±12 mL/min		±23 mL/min		±30 mL/min			
		Response time: 500 ms	Instantaneous flow rate	±3.2 mL/min		±2.9 mL/min		±3.8 mL/min		±7.2 mL/min		±9.5 mL/min		
Hysteresis				Variable										
Integrated unit display (displayed on controller)				0.1/1/10/100/1000/10,000 mL					0.01/0.1/1/10/100 L (standard), 0.1/1/10/100/1000/10000 mL (high resolution* ⁶)					
Display method				Status indicator										
Environmental resistance	Enclosure rating		IP65/IP67 (IEC60529), IP68G (JIS C0920)* ⁵											
	Ambient temperature		0 to 60°C (No freezing)				-10 to 60°C (No freezing)							
	Ambient humidity		35% to 85% RH (No condensation)											
	Vibration resistance		10 to 55 Hz, double amplitude 1.5 mm, 2 hours each for X,Y,Z direction											
	Shock resistance		50 G 11 ms, 3 times each for X,Y,Z direction											
Material	Sensor head			Head body: PPS/PPSU, in-cable amplifier: PPS, cable: PVC, controller connector: PPS/PBT/POM										
	Clamp set		For plastic pipe For metal pipe	Body, fixing screw: PPS, detection surface: special rubber, pipe support rubber: FKM, sensor head fixing screw: SUSXM7 Metal: SUS304/SUSXM7, detection surface: special rubber, clamp support rubber: FKM, sensor head fixing screw: SUSXM7										

*1 Liquid must allow for the passage of an ultrasonic pulse, as well as not contain large air pockets or excessive bubbles. Readings may become unstable depending on the type of the pipe.
 *2 The zero cut flow rate can be changed in the settings. When using the unit with a low flow rate range, perform an origin adjustment when the fluid is not moving if you change the zero cut flow rate.
 *3 This specification is valid when the flow velocity distribution is stable. This value does not take into account the effects of pulsation or fluctuations in flow velocity distribution due to facility factors. Convert the F.S. (full scale value) listed in the table according to the rated flow range. *4 The longer the response time is set, the more repeatability is improved. As a guideline, use $\sqrt{}$ (50 ms/response time) times.
 *5 The connector part of the sensor head cable is IP65/IP67. *6 Only controllers with serial numbers beginning with "G" (FD-XA1/XA2/XA5) are supported.

■ Shot amount repeatability (Typical values) * Plastic pipe/tube attachment

Sensor head model			FD-XS1		FD-XS8			FD-XS20			
Clamp set model			FD-XC1R1	FD-XC1R2	FD-XC8R1	FD-XC8R2	FD-XC8R3	FD-XC20R1	FD-XC20R2	FD-XC20R3	FD-XC20R4
Shot time	50 ms		±0.005 mL		±0.003 mL		±0.004 mL	±0.006 mL		±0.008 mL	
	1 s		±0.015 mL		±0.008 mL		±0.012 mL	±0.019 mL		±0.023 mL	
	10 s		±0.044 mL		±0.024 mL		±0.036 mL	±0.057 mL		±0.071 mL	

*1 Repeatability of the shot amount is the typical value for water, response time of 50 ms, no zero cut flow rate setting and after origin adjustment.
 *2 Variations due to facility factors (such as pulsation, valve control, liquid pooling, change in flow velocity distribution) are not taken into account in this value.

■ Controller (Standard)

Model		FD-XA1	FD-XA2	FD-XA5
Type		DIN rail type, main unit		Panel type, main unit
Display method		Output indicator, 4-digit 7 segment display, OLED, Stability level display		
Display refresh frequency		Instantaneous flow: approx. 5 times/second, Discharge amount/Accumulated flow approx. 30 times/second		
Response time		50 ms/ 100 ms/ 500 ms/ 1 s/ 2.5 s/ 5 s/ 10 s/ 30 s/ 60 s (selectable, default: 500 ms)		
Integration data storage interval		Written to the memory every 10 seconds		
Memory back up*1		EEPROM (data storage period: more than 10 years, number of data rewritable times: 1 million times or more)		
Detection mode (selectable)	ch.1	Instantaneous flow rate mode/Area mode/Pulse output (+) mode/Integrated flow mode/Shot mode		
	ch.2	Instantaneous flow rate mode/Area mode/Pulse output (-) mode/Shot mode/Error output mode/Bubble alert mode/Error + bubble alert mode		
Input/output	Output ch.1/2	NPN/PNP setting switch Open collector output: 30 V or lower, main unit: 50 mA or lower/ch.*2/expansion unit: 20 mA or lower/ch., residual voltage: 2 V or lower		
	Analogue output	4-20 mA/0-20 mA (selectable) load resistance: 500 ohms or lower	—	4-20 mA/0-20 mA (selectable) load resistance: 500 ohms or lower
	External input 1/2	Flow rate zero input/shot sampling input/integrated flow reset input/zero shift input (selectable) Short circuit current: NPN 1 mA or lower/PNP 2 mA or lower, input time: 20 ms or longer		
Network support		IO-Link*3	Supports NU Series	IO-Link*3
Power source	Power supply voltage	20 to 30 VDC including 10% ripple (P-P), Class 2		
	Current consumption	195 mA or lower (including the sensor head, excluding the load current)	185 mA or lower (including the sensor head, excluding the load current)	195 mA or lower (including the sensor head, excluding the load current)
Protection circuit		Power supply reverse connection protection, power surge protection, output short circuit protection, output surge protection		
Addition of expansion units		Up to 7** per main unit		—
Environmental resistance	Ambient temperature	-10 to +50°C (No freezing)		
	Ambient humidity	35% to 85% RH (No condensation)		
	Vibration resistance	10 to 55 Hz, double amplitude 1.5 mm, 2 hours each for X,Y,Z direction		
	Shock resistance	100 m/s ² (approx. 10 G) 16 ms pulse, 1000 times each for X,Y,Z direction		
Material		Main body case/front sheet: PC, Key top: POM, Cable: PVC		

*1 Internal data for full time recording can be read by USB (Ver.2.0) communication. *2 20 mA or lower/ch when adding expansion units. *3 IO-Link: Specification v1.1/COM2 (38.4 kbps) is supported. If the end of the cable needs to be an M12 connector when supporting IO-Link communication, connect an M12 conversion connector (OP-88296) to the cable. *4 Consult the manual for the number of serially-connectable devices to the N-bus.

■ FD-XE (For clog detection)

Sensor head model			FD-XS1E			FD-XS8E			FD-XS20E					
Supported pipe materials			Metal pipes, Plastic pipes (soft/hard)*1											
Supported fluids			Liquids (water, oil, adhesive, grease, chemical solution, etc.)*1											
Supported fluid temperature (Pipe surface temperature)			0°C (no freezing on the pipe surface) to 100°C											
Supported diameter	Plastic pipe/tube attachment	Clamp set model	FD-XC1R1		FD-XC1R2	FD-XC8R1	FD-XC8R2	FD-XC8R3	FD-XC20R1	FD-XC20R2		FD-XC20R3	FD-XC20R4	
		Outer diameter of pipe	ø3	1/8" (3.18 mm)	ø4	ø6	1/4" (6.35 mm)	ø8	3/8" (9.53 mm)	ø10		ø12	1/2" (12.7 mm)	
		Attachable range	ø2.7 to 3.7		ø3.5 to 4.5	ø5.5 to 6.5	ø5.9 to 6.9	ø7.5 to 8.5	ø9.0 to 10.0	ø9.5 to 10.5		ø11.5 to 12.5	ø12.2 to 13.2	
	Metal pipe attachment	Clamp set model	FD-XC1M			FD-XC8M			FD-XC20M1			FD-XC20M2		
		Outer diameter of pipe	ø3	1/8" (3.18 mm)	ø4	ø6	1/4" (6.35 mm)	ø8	3/8" (9.53 mm)	ø10	ø10.5	ø12	1/2" (12.7 mm)	ø13.8
		A designation	—	—	—	—	—	—	—	—	6 A	—	—	8 A
		Attachable range	ø2.8 to 5.5			ø5.5 to 8.3			ø8.3 to 10.8			ø10.8 to 14		
Maximum rated flow rate			1000 mL/min			3000 mL/min		8000 mL/min	15.00 L/min			20.00 L/min		
Zero cut flow rate*2 (variable, default)			50 mL/min			50 mL/min			0.15 L/min					
Display resolution (Displayed on controller)	Instantaneous flow		1/10 mL/min					0.01/0.1 L/min						
Repeatability			Response time 50 ms: ±20% of RD Response time 500 ms: ±15% of RD											
Hysteresis			Variable											
Display method			Status indicator											
Environmental resistance	Enclosure rating		IP65/IP67 (IEC60529), IP68G (JIS C0920)*3											
	Ambient temperature		0 to 60°C (No freezing)				-10 to 60°C (No freezing)							
	Ambient humidity		35% to 85% RH (No condensation)											
	Vibration resistance		10 to 55 Hz, double amplitude 1.5 mm, 2 hours each for X, Y, Z direction											
	Shock resistance		50 G 11 ms 3 times each for X,Y,Z direction											
Material	Sensor head		Head body: PPS/PPSU, relay amplifier: PPS, cable: PVC, controller connector: PPS/PBT/POM											
	Clamp set	For plastic pipe/tube	Body, fixing screw: PPS, detection surface: special rubber, pipe support rubber: FKM, sensor head fixing screw: SUSXM7											
		For metal pipe	Metal: SUS304/SUSXM7, detection surface: special rubber, clamp support rubber: FKM, sensor head fixing screw: SUSXM7											

*1 Liquid must allow for the passage of an ultrasonic pulse, as well as not contain large air pockets or excessive bubbles. Readings may become unstable depending on the type of pipe.

*2 The zero cut flow rate can be changed in the settings. When using the unit with a low flow rate range, perform an origin adjustment when the fluid is not moving if you change the zero cut flow rate.

*3 The connector part of the sensor head cable is IP65/IP67.

■ Controllers (For clog detection)

Model		FD-XA1E	FD-XA2E	FD-XA5E
Type		DIN rail type, main unit	DIN rail type, expansion unit	Panel type, main unit
Display method		Output indicator, 4-digit 7 segment display, OLED, Stability level display		
Display refresh frequency		Approximately 5 times/second		
Response time		50 ms/100 ms/500 ms (selectable, default: 500 ms)		
Memory back up		EEPROM (data storage period: more than 10 years, number of data rewritable times: 1 million times or more)		
Detection mode	ch.1	Instantaneous flow rate mode		
	ch.2	Error output mode		
Input/output	Output	NPN/PNP selectable Open collector output: 30 V or lower, main unit: 50 mA or lower/ch.*1/expansion unit: 20 mA or lower/ch., residual voltage: 2 V or lower		
	External input	Flow rate zero input/zero shift input (switchable) Short circuit current: NPN 1 mA or lower/PNP 2 mA or lower, input time: 20 ms or longer		
Network support		—	Supports NU Series	—
Power source	Power supply voltage	20 to 30 VDC including 10% ripple (P-P), Class 2		
	Current consumption	185 mA or lower (including the sensor head, excluding the load current)		
Protection circuit		Power supply reverse connection protection, power surge protection, output short circuit protection, output surge protection		
Addition of expansion units		Up to 7*2 per main unit		—
Environmental resistance	Ambient temperature	-10 to +50°C (No freezing)		
	Ambient humidity	35% to 85% RH (No condensation)		
	Vibration resistance	10 to 55 Hz, double amplitude 1.5 mm, 2 hours each for X,Y,Z direction		
	Shock resistance	100 m/s ² (approx. 10 G) 16 ms pulse, 1000 times each for X,Y,Z direction		
Material		Main body case/front sheet: PC, Key top: POM, Cable: PVC		

*1 20 mA or lower/ch when adding expansion units.

*2 Consult the manual for the number of serially-connectable devices to the N-bus.

Multi-output unit

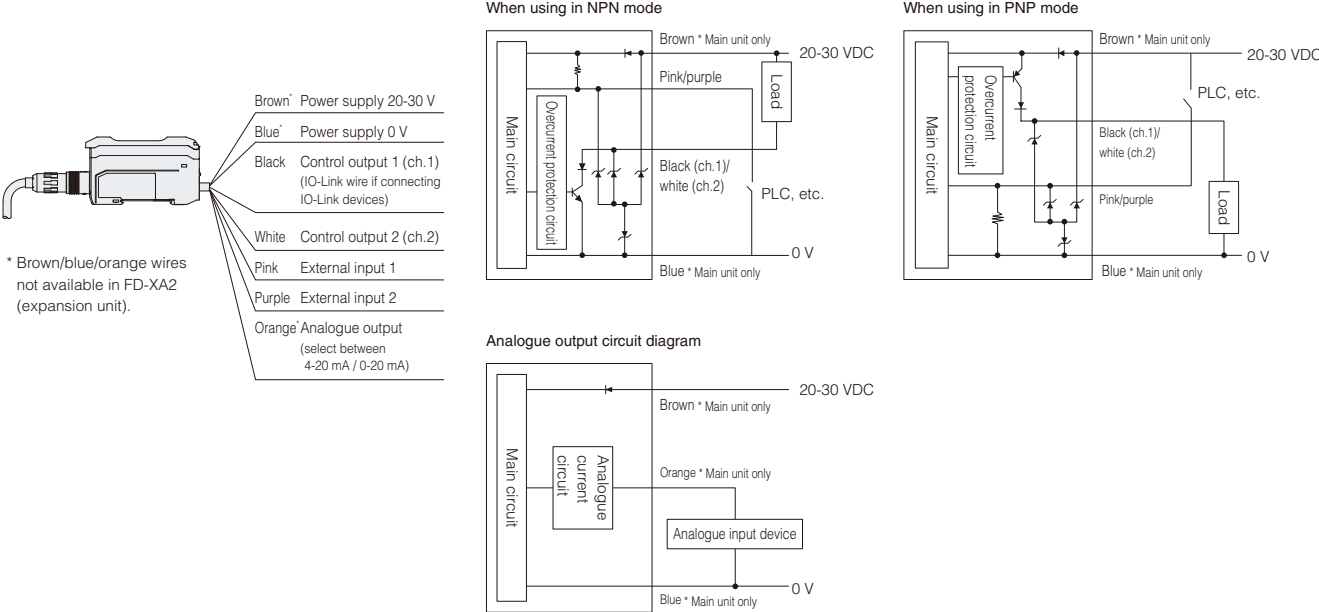
Model	NPN output	FS-MC8N
	PNP output	FS-MC8P
No. of inputs/outputs	Separate control outputs: 8, common output: 1, common input: 1	
Response time	Depends on the response time settings of the connected expansion units	
Unit expansion	Up to 8 expansion units can be connected. (However, each dual output type will be treated as 2 expansion units.) Allowable passing current: 1200 mA or less	
Indicators	STATUS indicator (green and red two-colour display), MEMORY indicator (orange), LOCK indicator (orange)	
Separate control outputs	NPN output	NPN open-collector, 30 V or less, 20 mA or less per output, residual voltage: 1.4 V or less
Common output	PNP output	PNP open-collector, 30 V or less, 20 mA or less per output, residual voltage: 1.6 V or less
External input time	Input time of the connected expansion units + 11 ms	
Protection circuit	Protection against reverse power connection, reverse output connection, output overcurrent, and output surge	
Power supply	Power voltage*1	10 to 30 VDC (including ripple (P-P) 10% or less), class 2 or LPS
	Power consumption*2	690 mW or less (when used as a solitary unit) (26 mA or less at 24 V/38 mA or less at 12 V [excluding the load current])
Environmental resistance	Ambient temperature	-20°C to +55°C (no freezing)
	Vibration resistance	10 to 55 Hz, double amplitude 1.5 mm; 2 hours each for X, Y, and Z axes
	Shock resistance	500 m/s ² , 3 times each in X, Y, and Z axes
Case material	Main unit and cover: polycarbonate	
Weight	Approx. 110 g	

*1 Match the rated power supply voltage of the expansion units to be connected to expand the system.
*2 The power consumption including the loads when the maximum number of units are connected is 38 W max.

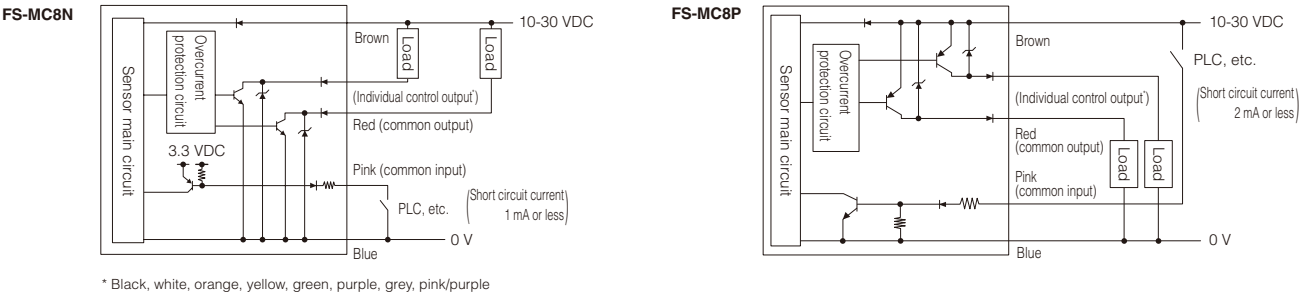
I/O Circuit Diagrams

Controller (Standard)

FD-XA1/XA2/XA5

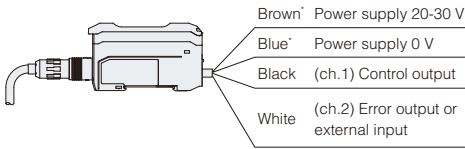


Multi-output unit



■ Controller (For clog detection)

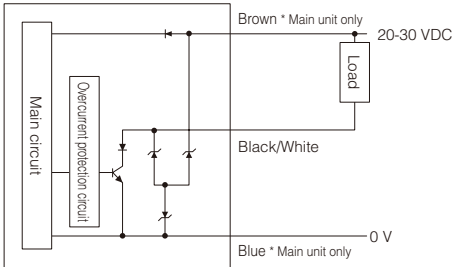
FD-XA1E/XA2E/XA5E



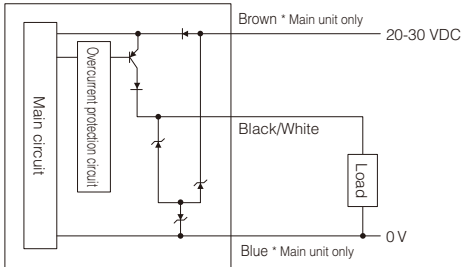
* No brown or blue wire in FD-XA2E (expansion unit).

Selected ch.2 function = output

When using in NPN mode

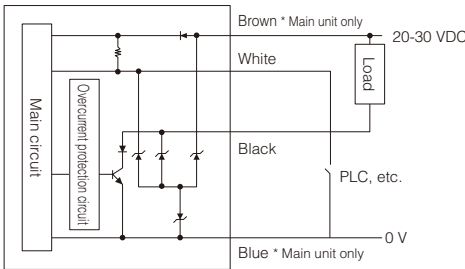


When using in PNP mode

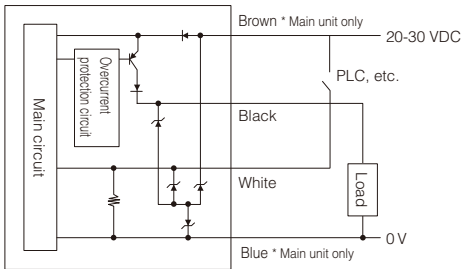


Selected ch.2 function = input

When using in NPN mode



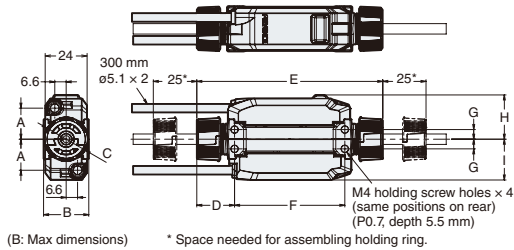
When using in PNP mode



Dimensions

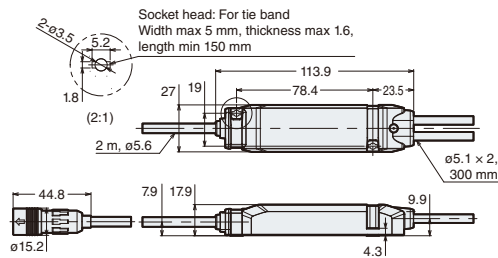
■ Sensor head

FD-XS1/XS8/XS20/XS1E/XS8E/XS20E + plastic piping clamp set

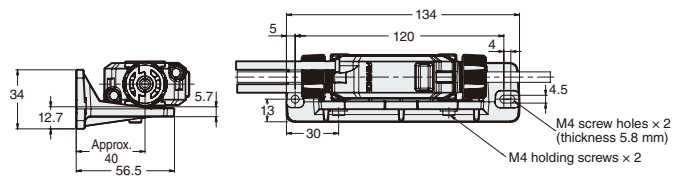


Sensor head	Clamp set	A	B	C	D	E	F	G	H	I
FD-XS1	FD-XC1R1	14.6	24.6	ø22	21.3	106	63.4	4	22.2	20.2
	FD-XC1R2	15	24.6	ø22	21.3	106.5	63.9	4.4	22.6	20.6
FD-XS8	FD-XC8R1	17.9	26	ø25.5	21.4	106.1	63.4	5.6	25.5	23.5
	FD-XC8R2	18	26	ø25.5	21.4	106.3	63.6	5.7	25.6	23.6
	FD-XC8R3	18.9	26	ø25.5	21.4	107.3	64.6	6.6	26.5	24.5
FD-XS20	FD-XC20R1	22.4	30	ø29.5	21.5	112.8	69.9	7.3	30	28
	FD-XC20R2	22.7	30	ø29.5	21.5	113.3	70.4	7.6	30.3	28.3
	FD-XC20R3	23.7	30	ø29.5	21.5	114.4	71.5	8.6	31.3	29.3
	FD-XC20R4	24	30	ø29.5	21.5	114.8	71.9	8.9	31.6	29.6

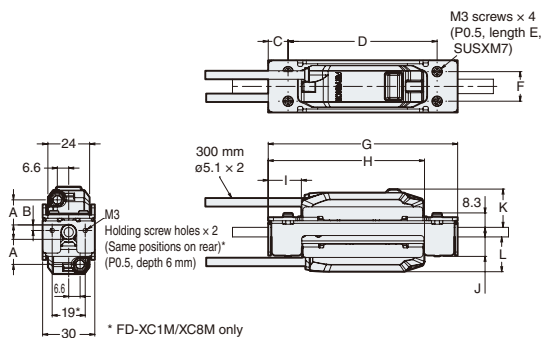
Relay amplifier



FD-XS1/XS8/XS20 securing bracket installed (optional, sold separately, OP-88294)

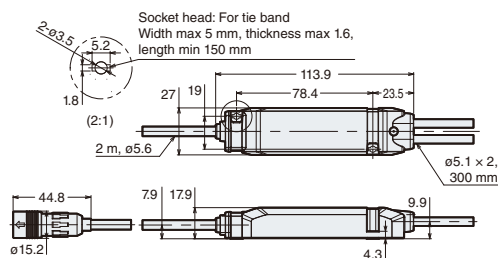


FD-XS1/XS8/XS20/XS1E/XS8E/XS20E + metal piping clamp set

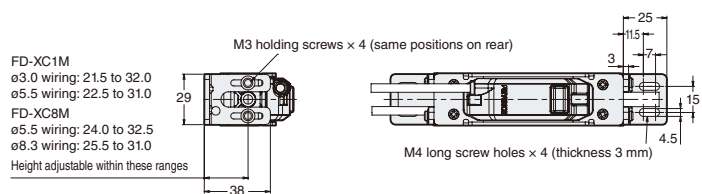


Sensor head	Clamp set	A	B	C	D	E	F	G	H	I	J	K	L
FD-XS1	FD-XC1M	12.7	2.2	10.8	79.9	20	16	102.5	83.5	22	10.7	20.3	18.3
FD-XS8	FD-XC8M	14.4	4.2	11.4	85.7	23	17	108.9	89.4	19.1	11.2	22	20
FD-XS20	FD-XC20M1	17.3	—	11.9	95.8	26	17.5	119.5	94.4	19.7	11.4	24.9	22.9
	FD-XC20M2	17.3	—	11.4	98.8	30	18	121.5	93.9	19.2	12.9	24.9	22.9

Relay amplifier

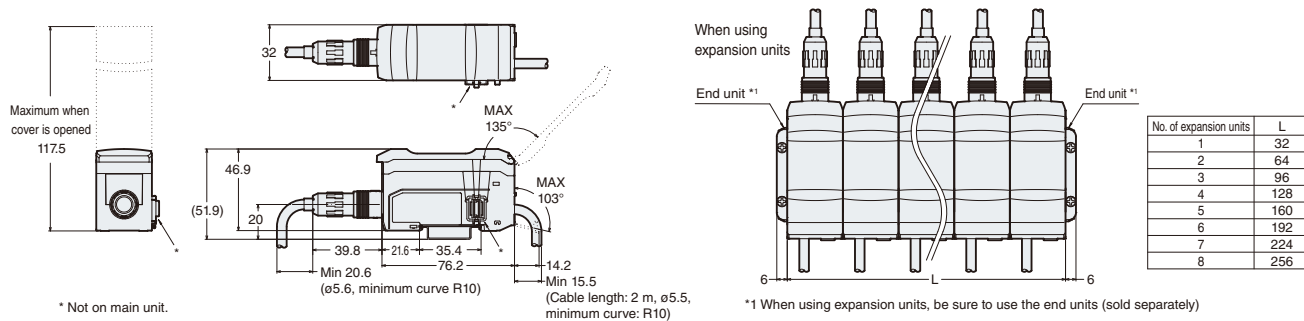


FD-XS1/XS8 securing bracket installed (optional, sold separately, OP-88297)

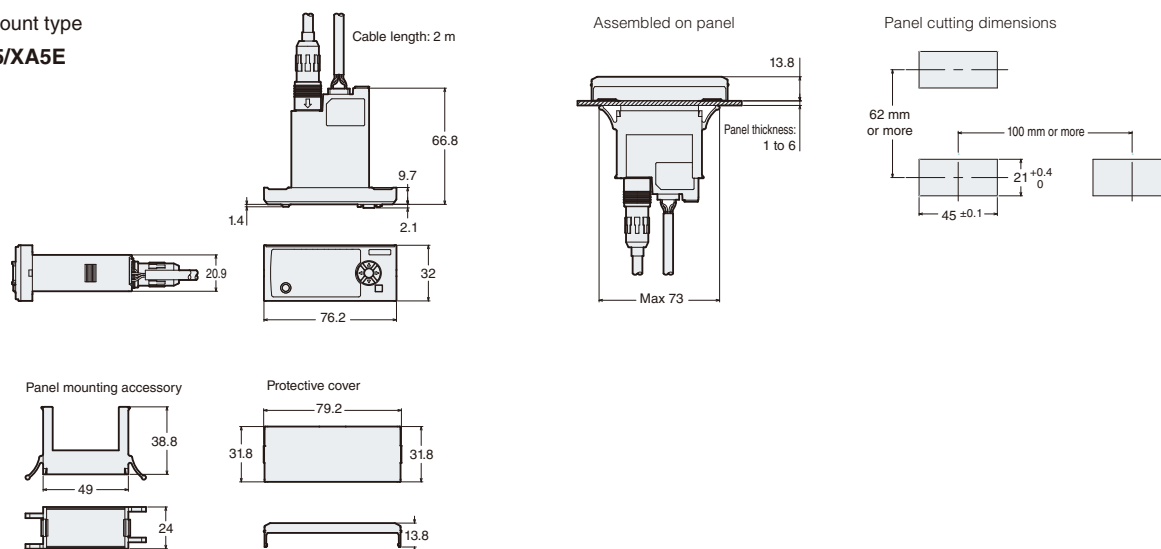


Controller

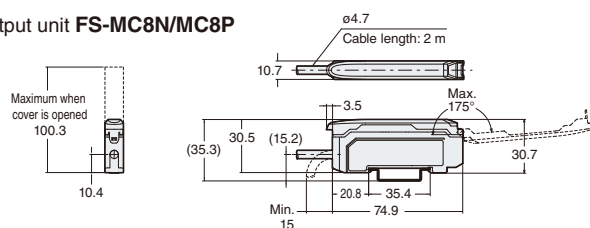
DIN rail mount type **FD-XA1/XA1E** (main unit) **/XA2/XA2E** (expansion unit)



Panel mount type **FD-XA5/XA5E**



Multi-output unit **FS-MC8N/MC8P**



Optional cables

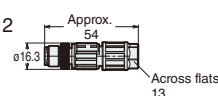
Sensor head-Controller extension cable

OP-88292/88293



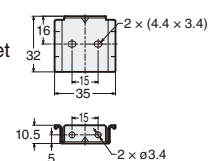
Loose wires-M12 adapter connector

OP-88296



DIN amplifier securing bracket (for main unit)

OP-88311





Clamp-on for micro flow rate monitoring



Please visit: www.keyence.com



SAFETY INFORMATION

Please read the instruction manual carefully in order to safely operate any KEYENCE product.

GLOBAL NETWORK

CONTACT YOUR NEAREST OFFICE FOR RELEASE STATUS

AUSTRIA
Phone: +43 2236 378266 0

BELGIUM
Phone: +32 15 281 222

BRAZIL
Phone: +55-11-3045-4011

CANADA
Phone: +1-905-366-7655

CHINA
Phone: +86-21-5058-6228

CZECH REPUBLIC
Phone: +420 220 184 700

FRANCE
Phone: +33-1-56-37-78-00

GERMANY
Phone: +49-6102-3689-0

HONG KONG
Phone: +852-3104-1010

HUNGARY
Phone: +36 1 802 73 60

INDIA
Phone: +91-44-4963-0900

INDONESIA
Phone: +62-21-2966-0120

ITALY
Phone: +39-02-6688220

JAPAN
Phone: +81-6-6379-2211

KOREA
Phone: +82-31-789-4300

MALAYSIA
Phone: +60-3-7883-2211

MEXICO
Phone: +52-55-8850-0100

NETHERLANDS
Phone: +31 40 20 66 100

PHILIPPINES
Phone: +63-2-8981-5000

POLAND
Phone: +48 71 36861 60

ROMANIA
Phone: +40 269 232 808

SINGAPORE
Phone: +65-6392-1011

SLOVAKIA
Phone: +421 2 5939 6461

SLOVENIA
Phone: +386 1 4701 666

SWITZERLAND
Phone: +41 43 455 77 30

TAIWAN
Phone: +886-2-2721-8080

THAILAND
Phone: +66-2-369-2777

UK & IRELAND
Phone: +44 1908-696-900

USA
Phone: +1-201-930-0100

VIETNAM
Phone: +84-24-3772-5555